SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F71BF

Life Insurance Mathematics 2

Semester 2 – 2015/16

Duration: Two Hours

You should answer all 7 questions.

Total: 60 marks

You may use any permitted calculator.

You should be provided with the yellow
"Formulae and Tables for Actuarial Examinations".

Question 1

A life office issues a large number of 10-year non-profit endowment assurance policies to males aged exactly 55.

The sum assured under each policy is £50,000, payable at the date of maturity or at the end of the year of earlier death.

Level premiums are payable annually in advance throughout the term of the policy.

The premium basis is as follows-

Mortality: AM92 select
Interest: 6% per year
Expenses: £300 at the start of each year.

The reserving basis is as follows-

Net premium method
Mortality: AM92 ultimate
Interest: 4% per year

(i) Show that the annual premium is slightly more than £4,000. *[2 marks]*

(ii) Calculate the policy values at durations 0, 2 and 10 years. *[5 marks]*

[Total 7 marks]

Question 2

A temporary annuity with term n years is issued to a life age x. The annuity is payable continuously at a rate of £t per year, where t is the length of time since the start of the annuity.

The annuity stops at time n, or at the death of the life, whichever is first.

The force of interest is a constant = δ. The force of mortality is variable, μ_{x+t}.

(i) State Thiele's differential equation for the contract. You are not required to prove your equation. *[2 marks]*

(ii) Give the boundary condition at time n. *[1 mark]*

(iii) Use Euler's method to find $V(n-h)$ and $V(n-2h)$, where h is small. *[4 marks]*

[Total 7 marks]

Question 3

A whole-life critical illness scheme, issued to a life aged x, pays a lump sum of £150,000 if a critical illness is diagnosed. If there is no such diagnosis, the scheme pays a lump sum of £80,000 on death.

(i) Draw and label the transition diagram for this scheme. You should define the transition intensities, and state any required assumptions. *[4 marks]*

(ii) Write an expression for the expected present value of this benefit. *[3 marks]*

[Total 7 marks]

Question 4

A pension scheme has a member now aged 72, who has a wife aged 68. The scheme pays him a pension of £12,000 per year for life. If his wife is still alive when he dies, she will be entitled to a pension of £9,000 per year for the rest of her life. Pensions are paid continuously.

Calculations should be carried out on the following basis-

Interest: 4% per year

Mortality: PA92C20

Pension increases: none

(i) Calculate the expected present value of the member's pension. *[2 marks]*

(ii) Calculate the expected present value of the widow's pension. *[3 marks]*

(iii) Explain briefly what is meant by the "20" in PA92C20. *[1 mark]*

[Total 6 marks]

Question 5

The workers in a manufacturing company are subject to two modes of decrement: mortality and withdrawal from employment.

Forces of mortality and withdrawal are constant between integer ages. The forces of mortality and withdrawal for employees aged 40 to 42 are given in the following table:

Age	μ	w
40-41	0.002	0.05
41-42	0.003	0.04

(i) Out of 100,000 lives aged 40 exact, how many will still be in service at age 41 exact? *[3 marks]*

(ii) Out of 100,000 lives aged 40 exact, how many will die between age 41 exact and age 42 exact? *[4 marks]*

[Total 7 marks]

Question 6

A life insurance company issues a three-year unit-linked endowment assurance policy to 1000 lives aged 50 exact. Level premiums of £4,000 are payable annually in advance throughout the term of the policy or until earlier death. The premium allocation rate (%) at time *t* is given by:
[66 + 30*t*] where *t* = 0, 1 and 2.

The units are subject to a bid-offer spread of 5%. An annual management charge of 0.80% of the bid value of units is deducted at the end of each policy year.

Management charges are deducted from the unit fund before any death, surrender or maturity benefits are paid.

If the policyholder dies during the term of the policy, the death benefit of £12,000 or the bid value of the units if higher, is payable at the end of the policy year of death.

The policyholder may surrender the policy only at the end of each policy year. On surrender at the end of the policy year or on survival to the end of the term, the current bid value of the units is payable.

The company uses the following assumptions in carrying out profit tests of this contract:

Rate of growth on assets in the unit fund	8% per annum
Rate of interest on non-unit fund cash flows	3% per annum
Mortality	q= 0.01 at all ages
Surrender	15% at the end of first, second and third policy years only
Initial expenses	£400
Renewal expenses	£100 per annum on the second and subsequent premium dates
Initial commission	4% of first premium
Renewal commission	2% of the second and subsequent years' premiums
Risk discount rate	10% per annum

[Question 6 continues on the next page]

Question 6 (continued)

(i) Calculate the expected maturity benefit. *[5 marks]*

(ii) Calculate the expected present value of the profit for the policy on the assumption that the company does not zeroise future expected negative cash flows. *[10 marks]*

(iii) Suppose the company sets up reserves in order to zeroise future negative expected cash flows.

Calculate the expected present value of the profit for the policy allowing for the cost of setting up these reserves. *[5 marks]*

[Total 20 marks]

Question 7

The following table gives a summary of mortality for mineworkers in a country, and for the country as a whole.

	Mineworkers		Whole country	
Age group	Population	Deaths	Population	Deaths
16-30	30,000	200	4 million	10,000
31-49	12,000	160	4 million	30,000
50-65	3,000	200	3 million	100,000
Total	45,000	560	11 million	140,000

(i) Calculate the crude death rate for mineworkers. *[1 mark]*

(ii) Calculate the crude death rate for the whole country. *[1 mark]*

(iii) Calculate the standardised mortality ratio for the mineworkers, using the whole country as the standard population. *[3 marks]*

(iv) Explain why the standardised mortality ratio might be preferred to crude death rates. *[1 mark]*

[Total 6 marks]

END OF PAPER